Accelerating Polynomial System Solving via Factorization and Unsatisfiable-Core Learning

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August 24, 2026

Abstract

Polynomial systems are a fundamental language for describing algebraic and geometric problems across mathematics and its applications. Beyond finding individual solutions, one often seeks global properties of the solution space, such as its dimension and irreducible components, or structural properties and decompositions of the underlying polynomial ideal. For large polynomial systems with many variables and high degrees, determining these global properties can become intractable. Leading approaches include variants of Buchberger's algorithm with syzygy, decomposition techniques, resultants, and cylindrical algebraic decomposition. However, even the most modern approaches still often fail on practical problems or slight perturbations of standard benchmarks. Here we introduce a framework that divides a polynomial system into simpler branches. It assumes only that the branch solver describes satisfiable branches and certifies unsatisfiable (UNSAT) ones. If the solver also returns an UNSAT core, the framework uses it to prune later branches, following the same basic idea as conflict-driven clause learning (CDCL). In our experiments, we use Gröbner-basis solvers and apply the method to two systems arising from research in quantum field theory for which no standard method in our comparison succeeds. Together, these ideas offer a path toward the exact analysis of previously inaccessible polynomial systems.

1 Introduction

Systems of multivariate polynomial equations appear whenever a symbolic model contains nonlinear algebraic constraints. Applications include robotics, coding theory, optimization, statistics, and physics [25]. In many of these problems, an isolated

solution is not enough. One needs the dimension of the solution set, its irreducible components, or exact relations that hold on every solution. These are questions about the variety and its defining ideal, rather than about a single point [7, 15].

Elimination by resultants provided an early systematic way to attack polynomial systems, but the introduction of Gröbner bases gave a common algorithmic framework for elimination, ideal membership, dimension, and related tasks [7, 15]. Buchberger developed this framework in his 1965 thesis [6]. Later algorithms changed how the expensive reductions are organized. F4 replaces individual polynomial reductions with sparse linear algebra, while F5 uses signatures to avoid many reductions to zero [9, 10]. Primary-decomposition algorithms use Gröbner bases and factorization to separate components [12]. Numerical algebraic geometry takes a different route: homotopy continuation and witness sets provide isolated roots numerically [23, 24].

There are recent works that try to use machine learning methods to solve polynomial systems of equations. Peifer et al. trained a reinforcement-learning policy for pair selection in Buchberger's algorithm [20]. Kera et al. trained transformers to compute Gröbner bases directly for a restricted class of zero-dimensional radical ideals [14].

Despite the progress in the field, larger systems remain inaccessible to current methods.

Our framework divides a polynomial system into simpler systems by factorizing the polynomials. The systems we get are the minimal hitting sets of factors, that is a set of factors that include a factor from each equation. The output is an exact collection of branch descriptions for the full solution set, including positive-dimensional families.

The framework uses UNSAT core learning to prune hitting sets that include a set of unsatisfiable constraints, adapting ideas from polynomial UNSAT-core extraction and conflict-driven learning [26, 18, 17].

Our framework does not require a particular branch solver or UNSAT core-extraction procedure.

Some factor branches can still be too expensive for the chosen subsolver. After a timeout, we branch again splitting a selected variable into the cases $x = 0$ and $x \neq 0$. Assuming the value of x should help the subsolver.

We prove that our methods preserve the complete solution set for any branch solver satisfying the stated conditions. We compare the resulting implementation with direct Gröbner-basis computation, symbolic decomposition, and numerical homotopy on generated benchmarks. We then apply it to two systems from quantum field theory for which the standard methods in our comparison do not find any solutions.

2 Preliminaries and notation

2.1 Polynomial systems, ideals, and varieties

Let K be a field, let $\overline{K}$ be its algebraic closure, and write $R = K[x_1, \ldots, x_n]$. The input is a finite system $P = \{p_1, \ldots, p_m\} \subseteq R$ of nonzero polynomials. Unless stated otherwise, a solution is a point in $\overline{K}^n$. For a finite set $Q \subseteq R$, its solution set is

$$V(Q) = \{a \in \overline{K}^n : q(a) = 0 \text{ for every } q \in Q\}.$$

We call Q satisfiable if $V(Q) \neq \emptyset$ and unsatisfiable (UNSAT) if $V(Q) = \emptyset$.

The ideal generated by Q, denoted $\langle Q \rangle$, consists of all sums $\sum_{q \in Q} h_q q$ with $h_q \in R$.

For $X \subseteq \overline{K}^n$, its vanishing ideal $I(X)$ contains every polynomial in R that vanishes on all of X. The radical of an ideal J is $\sqrt{J} = \{g \in R : g^r \in J \text{ for some } r \geq 1\}$. Hilbert's Nullstellensatz gives

$$I(V(Q)) = \sqrt{\langle Q \rangle}.$$

Thus $V(Q) = \emptyset$ exactly when $\langle Q \rangle = \langle 1 \rangle$. These facts connect solution sets with computations in polynomial ideals [7, 15].

2.2 Monomial orders and Gröbner bases

Let $\mathbb{M}_n = \{x^\alpha : \alpha \in \mathbb{N}^n\}$ be the monomials of R. A monomial order $\prec$ is a total order on $\mathbb{M}_n$ such that every nonempty subset has a least element and

$$u \prec v \quad \Longrightarrow \quad uw \prec vw \qquad (u, v, w \in \mathbb{M}_n).$$

For a nonzero polynomial p, the leading monomial $\mathrm{LM}(p)$ is the greatest monomial of p under $\prec$.

For an ideal J, define its leading-monomial ideal by

$$\mathrm{LM}(J) = \langle \mathrm{LM}(f) : f \in J \setminus \{0\} \rangle.$$

A finite set $G \subseteq J$ is a Gröbner basis of J if

$$\mathrm{LM}(J) = \langle \mathrm{LM}(g) : g \in G \rangle.$$

$p \in J$ if and only if its division by G has a 0 remainder, so division decides ideal membership.

Every ideal has a unique reduced Gröbner basis for a fixed monomial order. The reduced basis is $\{1\}$ if and only if $J = \langle 1 \rangle$. Therefore, a Gröbner basis computation can describe a branch and detect when it is UNSAT [7, 15].

An UNSAT core of a system Q is a subset $U \subseteq Q$ with $V(U) = \emptyset$. An UNSAT core is minimal if there is no $U' \subsetneq U$ such that $V(U') = \emptyset$. Any system containing U is also UNSAT, since adding equations cannot introduce solutions.

2.3 Factor sets and hitting sets

Fix a factorization of each input polynomial over K:

$$p_i = a_i \prod_{f \in C_i} f^{e_{i,f}},$$

where $a_i \in K \setminus \{0\}$, C_i contains the distinct nonconstant irreducible factors of p_i, and $e_{i,f} \geq 1$. Multiplicity does not affect the zero set. A point satisfies $p_i = 0$ exactly when at least one factor in C_i vanishes at that point.

A hitting set for $C_1, \ldots, C_m$ is a set $H \subseteq \bigcup_i C_i$ such that $H \cap C_i \neq \emptyset$ for every i. It is minimal if no proper subset is also a hitting set. We write $\mathcal{H}(P)$ for the family of minimal hitting sets obtained from P. Each $H \in \mathcal{H}(P)$ defines a branch in which all factors in H vanish [1].

2.4 Encoding nonzero variables

There are two standard encodings of nonzero variables as a polynomial equation. The first one encodes each $x \neq 0$ constraint as

$$\sigma_x = 1 - s_x x. \tag{1}$$

Here s_x is a new variable.

The second encoding combines the constraints $x \neq 0$ for all $x \in N$ into one equation

$$\sigma_N = 1 - s \prod_{x \in N} x. \tag{2}$$

For the first encoding, we project away the auxiliary variables s_x. For the second, we project away s. In both cases, the remaining points are the solutions of the original system that satisfy the nonzero conditions.

3 Branching framework

3.1 Framework description

Solving a system of polynomial equations is double exponential hard [19]. Thus breaking a system of polynomial equations into smaller subproblems allows for more efficient solving.

We break a system into smaller subproblems by iteratively doing two basic steps for each subproblem:

1. Interreduce the current subproblems.
2. Factor each generator and enumerate the minimal hitting sets of the factor sets.

Here is the pseudocode for the refinement of a polynomial system Q into a smaller set of subproblems.

Algorithm 1 Refinement

Require: A set of polynomials Q and A

Interreduce Q ▷ Optional
Factor each generator of Q
$\mathcal{H} \leftarrow$ the minimal hitting sets of the factor sets
$T \leftarrow \emptyset$
for all $H \in \mathcal{H}$ **do**
    if $H = Q$ or $H \in A$ **then**
        $T \leftarrow T \cup \{H\}$
    else
        $T \leftarrow T \cup \textsc{Refine}(H, A \cup \{\widetilde{Q}\})$
    end if
end for
return T

Let $\mathcal{L}(P)$ denote the terminal hitting sets obtained from P. For each $H \in \mathcal{L}(P)$, the branch solver must either describe $V(H)$ exactly or certify that $V(H) = \emptyset$.

The solver may also return a certified UNSAT core $U \subseteq H$, any later hitting set containing U can be determined to be UNSAT without calling the solver.

Algorithm 2 Branching

Require: A set of polynomials $P = \{p_1, \ldots, p_m\}$ and an exact branch solver $\mathcal{B}$.

$C \leftarrow \emptyset$ ▷ Certified UNSAT cores
$S \leftarrow \emptyset$ ▷ Solution-set representations
$R \leftarrow \textsc{Refine}(P, \emptyset)$
for all $H \in R$ **do**
    if H contains an UNSAT core from C **then**
        continue
    end if
    Use $\mathcal{B}$ to solve H
    if $\mathcal{B}$ returns a representation of $V(H)$ **then**
        Add the returned representation to S
    else
        if $\mathcal{B}$ returns a certified UNSAT core **then**
            Add the core to C
        end if
    end if
end for
return S, C

The order of the terminal hitting sets affects running time but not correctness. We use the number of generators in a hitting set as its estimated cost and process smaller sets first. An UNSAT core found in a small set may then prune larger sets before they reach the solver.

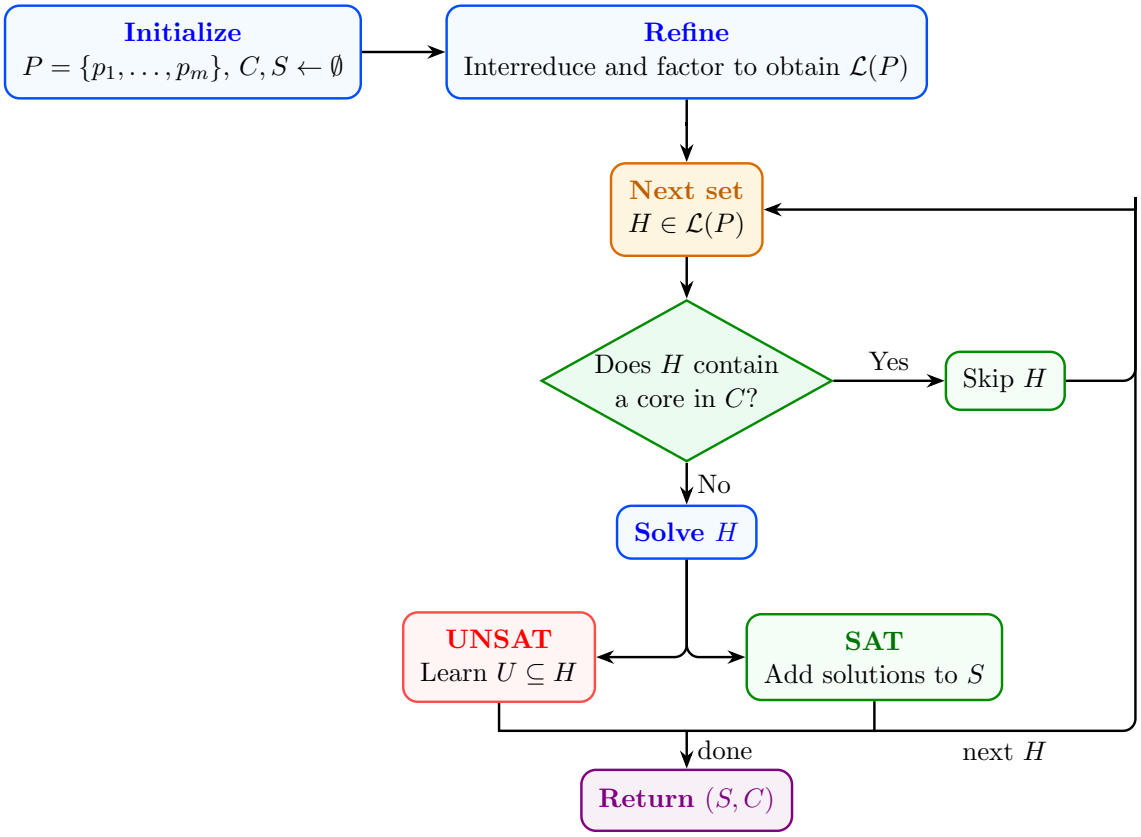


Figure 1: Flowchart of the branching procedure of Algorithm 2.

3.2 Theoretical analysis

Let S be a collection of solution-set representations, and $\mathrm{Sol}(S)$ be the union of the represented sets. To show that Algorithm 2 is sound and complete, we prove that it returns S with $\mathrm{Sol}(S) = V(P)$, and if the solver terminates then the algorithm terminates.

Lemma 1 (Correctness of interreduction) *Suppose that one generator p_i is replaced by*

$$p_i' = p_i - \sum_{j \neq i} a_j p_j, \qquad a_j \in K[x_1, \ldots, x_n].$$

Then

$$\langle p_1, \ldots, p_i, \ldots, p_m \rangle = \langle p_1, \ldots, p_i', \ldots, p_m \rangle.$$

Any finite sequence of these reductions therefore preserves the generated ideal and its zero set.

Proof 1 *The polynomial p_i' belongs to the original ideal, while $p_i = p_i' + \sum_{j \neq i} a_j p_j$ belongs to the ideal generated by the updated system. Each generating set therefore generates the other. The same argument applies at every step of a sequential interreduction.*

Lemma 2 (Sufficiency of minimal hitting sets) *Every hitting set T of the factor family contains a minimal hitting set $H \subseteq T$, with $V(T) \subseteq V(H)$. It is therefore enough to consider minimal hitting sets.*

Proof 2 *Because the factor sets are finite, repeatedly removing redundant factors from T eventually produces a minimal hitting set H. Since $H \subseteq T$, any point in $V(T)$ also lies in $V(H)$.*

For a system Q, write $\mathcal{H}(Q)$ for the minimal hitting sets obtained in one factorization round.

Theorem 1 (One-round branch decomposition) *The solution set of Q is*

$$V(Q) = \bigcup_{H \in \mathcal{H}(Q)} V(H).$$

Proof 3 *Write $Q = \{q_1, \ldots, q_m\}$ and let F_i be the nonconstant irreducible factors of q_i. Take $x \in V(Q)$. Since $q_i(x) = 0$, at least one factor in F_i vanishes at x for every i. The set of all factors that vanish at x is a hitting set of $F_1, \ldots, F_m$. Because the factor sets are finite, this set contains a minimal hitting set H. Every factor in H vanishes at x, so $x \in V(H)$.*

For the reverse inclusion, take $x \in V(H)$ for some $H \in \mathcal{H}(Q)$. The set H intersects every F_i, so each q_i has a factor that vanishes at x. Hence $q_i(x) = 0$ for every i, and $x \in V(Q)$.

Lemma 3 (Recursive refinement) *Suppose refinement of Q terminates with the finite collection $\mathcal{L}(Q)$. Then*

$$V(Q) = \bigcup_{H \in \mathcal{L}(Q)} V(H).$$

Proof 4 *Interreduction preserves the zero set by Lemma 1. Theorem 1 then replaces each nonterminal node by the union of its children. Apply this identity from the root to the leaves of the finite refinement tree. A branch with no hitting set has an empty zero set, while a repeated branch is retained as a terminal set. The remaining leaves are exactly* $\mathcal{L}(Q)$.

The recursive decomposition also gives an ideal identity.

Theorem 2 (Radical decomposition) *The radical of the input ideal satisfies*

$$\sqrt{\langle P \rangle} = \bigcap_{H \in \mathcal{L}(P)} \sqrt{\langle H \rangle}.$$

Proof 5 *By Lemma 3,* $V(P) = \bigcup_{H \in \mathcal{L}(P)} V(H)$. *Taking vanishing ideals gives*

$$I(V(P)) = I\left(\bigcup_{H \in \mathcal{L}(P)} V(H) \right) = \bigcap_{H \in \mathcal{L}(P)} I(V(H)).$$

Hilbert's Nullstellensatz identifies $I(V(P))$ *with* $\sqrt{\langle P \rangle}$ *and each* $I(V(H))$ *with* $\sqrt{\langle H \rangle}$, *which gives the stated equality.*

Lemma 4 *Let* U *be an UNSAT core. If* $U \subseteq H$, *then* $V(H) = \emptyset$.

Proof 6 *Since every point in* $V(H)$ *must satisfy every polynomial in* U, *we have* $V(H) \subseteq V(U)$. *The core* U *is UNSAT, so* $V(U) = \emptyset$ *and therefore* $V(H) = \emptyset$.

A smaller core is useful because more later branches may contain it.

For the following result, assume that refinement terminates and that the branch solver terminates on every unpruned terminal set H. The solver either returns $V(H)$ exactly or certifies that $V(H) = \emptyset$. Any core used for pruning must be a certified subset $U \subseteq H$.

Theorem 3 *Algorithm 2 is sound and complete for* P.

Proof 7 *We check branch construction, pruning, and solving in turn.*

Recursive branch construction. *Every interreduction round preserves the current ideal by Lemma 1. In each factorization round, Lemma 2 shows that nonminimal hitting sets add no solutions. Lemma 3 therefore gives*

$$V(P) = \bigcup_{H \in \mathcal{L}(P)} V(H).$$

Thus the terminal branch collection represents exactly $V(P)$.

Loop invariant. *Let* R *be the set of unprocessed branches and* S *the collection of solution-set representations returned so far. At the start of the loop,*

$$\mathrm{Sol}(S) \subseteq V(P) \qquad \textit{and} \qquad V(P) \subseteq \mathrm{Sol}(S) \cup \bigcup_{H \in R} V(H).$$

The first inclusion is the soundness invariant. The second is the completeness invariant: every solution outside $\mathrm{Sol}(S)$ *remains in a pending branch. Both hold initially because* $S = \emptyset$, $R = \mathcal{L}(P)$, *and the recursive branch decomposition holds.*

Core pruning. *Suppose the algorithm skips* H *because it contains a learned core* U. *Lemma 4 gives* $V(H) = \emptyset$. *Removing* H *from* R *therefore leaves the completeness invariant unchanged, so soundness is unchanged as well.*

Branch solving. *If* H *is satisfiable, the solver adds to* S *a representation of* $V(H)$. *Since* $V(H) \subseteq V(P)$, *the soundness invariant is preserved. Moving* $V(H)$ *from the pending union into* $\mathrm{Sol}(S)$ *also preserves the completeness invariant. If* H *is UNSAT, then* $V(H) = \emptyset$, *so removing it from* R *preserves both invariants.*

Return. *When the loop ends,* $R = \emptyset$. *The completeness invariant gives* $V(P) \subseteq \mathrm{Sol}(S)$, *while the soundness invariant gives* $\mathrm{Sol}(S) \subseteq V(P)$. *Hence* $\mathrm{Sol}(S) = V(P)$.

The theoretical analysis does not require K to be algebraically closed. If a system has no solution over the algebraic closure, it has no solution over K.

Now we shall show that the algorithm terminates.

Theorem 4 *If the exact solver is known to terminate then Algorithm 2 terminates.*

Proof 8 *First consider the recursion in* Refine. *If interreduction is disabled, every child consists of irreducible polynomials. Factoring such a child again returns the child itself, so it is terminal. Thus the refinement tree has finite depth in this case.*

Suppose now that interreduction is enabled. For a finite system Q, set

$$M(Q) = \langle \mathrm{LM}(q) : q \in Q \rangle.$$

A reduction of one generator modulo the others can only enlarge $M(Q)$: if its leading monomial changes, the old leading monomial was divisible by the leading monomial of a generator that remains in the system. After interreduction, the leading monomials of the nonzero generators are pairwise nondividing. If H is a hitting-set child, then for every $q \in Q$ some $h \in H$ divides q, and hence $\mathrm{LM}(h) \mid \mathrm{LM}(q)$. Consequently $M(Q) \subseteq M(H)$. The inclusion can be an equality only when every selected factor is associate to the generator it hits; after normalization this says $H = Q$, and the branch is terminal. Hence $M(Q)$ strictly increases on every recursive edge.

The polynomial ring is Noetherian, so its monomial ideals satisfy the ascending-chain condition. There can therefore be no infinite path in the refinement tree. Each node has finitely many children, since it has finitely many factors and finitely many minimal hitting sets. König's lemma now implies that the entire tree is finite. Consequently, $R = \textsc{Refine}(P, \emptyset)$ is computed in finite time and is finite.

Algorithm 2 therefore performs only finitely many loop iterations. A pruned branch requires no solver call, and on every unpruned branch the exact solver terminates by hypothesis. All remaining operations are finite, so the algorithm terminates.

4 Branching with variable zeroing

4.1 Branching with variable zeroing

Lets say that the solver doesn't terminate within reasonable amount of time for an hitting set given by Algorithm 2.

In this case, we can further simplify the system by branching further. In each branch assume some side condition holds, and letting Algorithm 2 run on the simplified system. This splitting can be done recursively until the solver terminates or the computation budget is exhausted.

For completeness, the union of the side conditions must cover all possibilities. A set of conditions is a good choice if it is helpful for your solver, and works well with algorithm 1.

For this work the branching conditions is whether a variable is zero or nonzero. This simplification choice is used in other works [**?**]. in this work we considered physical systems and determined through past experience that we expect a variable to be always zero, or always nonzero.

Algorithm 3 Recursive branching with variable zeroing

Require: A polynomial system P, and a timeout τ.

function ZeroingBranching(P, τ)
  $X \leftarrow$ Variables(P)
  return SolveByZeroing($P, X, \emptyset, \emptyset, \tau, \emptyset$)
end function
function SolveByZeroing(P, X, Z, N, τ, C)
  $(S, T) \leftarrow$ BranchingWithTimeout(P, τ, C) ▷ Representations inherit the path conditions (Z, N)
  $U \leftarrow \emptyset$ ▷ Branches left unresolved
  for all $Q \in T$ **do**
    if $X = \emptyset$ **then**
      $U \leftarrow U \cup \{(Q, Z, N)\}$
      continue
    end if
    $x \leftarrow$ ChooseVariable(Q, X)
    $Q_0 \leftarrow$ Substitute($Q, x, 0$)
    $Q_1 \leftarrow$ UpdateSaturation($Q, N \cup \{x\}$)
    $(S_0, U_0, C) \leftarrow$ SolveByZeroing($Q_0, X \setminus \{x\}, Z \cup \{x\}, N, \tau, C$)
    $(S_1, U_1, C) \leftarrow$ SolveByZeroing($Q_1, X \setminus \{x\}, Z, N \cup \{x\}, \tau, C$)
    $S \leftarrow S \cup S_0 \cup S_1$
    $U \leftarrow U \cup U_0 \cup U_1$
  end for
  return (S, U, C)
end function

For Algorithm 3 we use a global set C of learned UNSAT cores.

Let N be the variables declared nonzero on the current path. In Subsection 2.4 we described two ways to encode the nonzero conditions as a polynomial equations.

Solutions of the nonzero branch are projected onto the original variables, so the saturation variable does not appear in the output. The zero and nonzero decisions remain part of the solution-set representation.

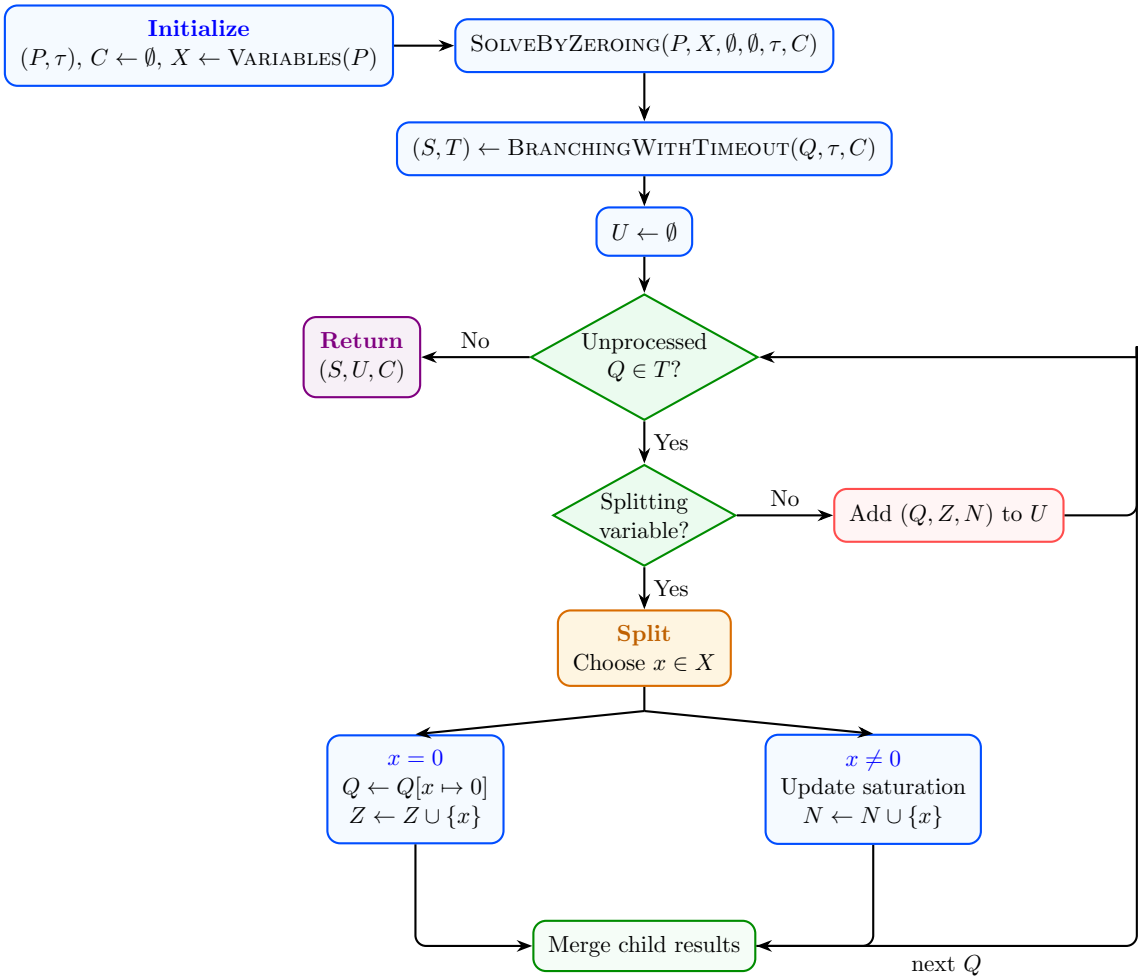


Figure 2: Flowchart of the recursive variable-zeroing procedure of Algorithm 3.

The run on `systems/system.txt`, with a five-second timeout for each hitting set, produces the recursion tree in Figure 3. The variable chosen after a timeout is written inside the corresponding hitting-set node. Its outgoing edges give the zero and nonzero cases.

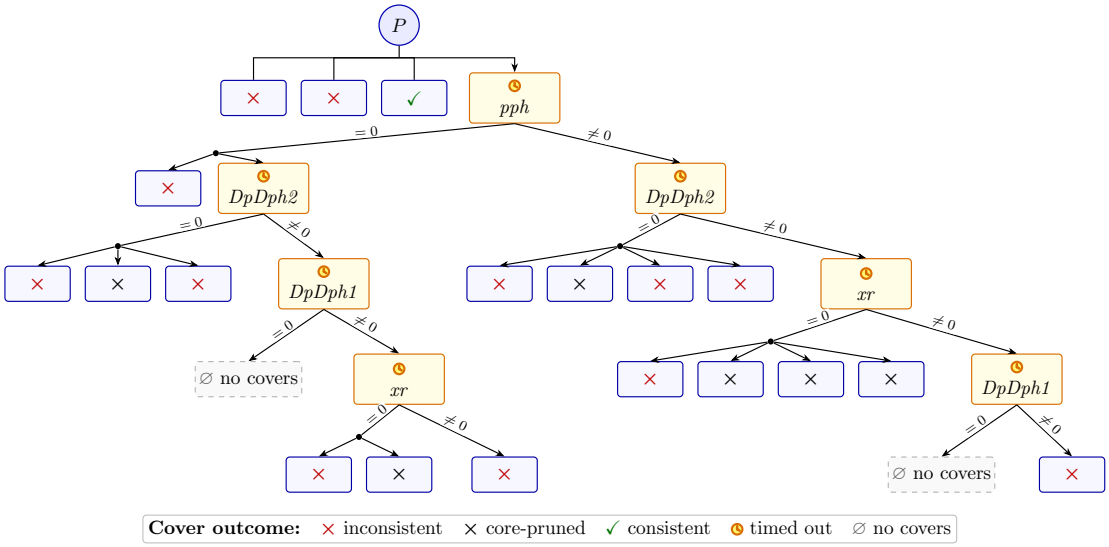


Figure 3: Recursion tree for `systems/system.txt`. Each hitting set is shown separately; ∅ marks a branch that produces no hitting sets.

4.2 Choice of variable for splitting

The choice of variable for splitting affects the efficiency of the algorithm. Every split creates at least two new hitting sets to check. If a variable barely occurs in the current hitting set, both recursive calls may repeat most of the same work. A variable that occurs in many generators is a more promising choice. Setting it to zero removes all terms containing it at once. In the nonzero branch, it is appended to the product in Equation (2).

For each candidate variable x that is still available on the current branch, define

$$\begin{aligned} g_Q(x) &= \#\{q \in Q : x \text{ occurs in } q\}, \\ t_Q(x) &= \#\{(q, m) : q \in Q,\ m \text{ is a term of } q,\ \deg_x(m) > 0\}, \\ e_Q(x) &= \sum_{q \in Q} \sum_{\substack{m \text{ a term of } q, \\ \deg_x(m) > 0}} \deg_x(m). \end{aligned}$$

Its score is the triple

$$h_Q(x) = \big(g_Q(x), t_Q(x), e_Q(x)\big),$$

and scores are compared lexicographically.

4.3 Branch contraction

Let $\mathcal{H}_0$ and $\mathcal{H}_1$ be the minimal hitting sets for $x = 0$ and $x \neq 0$. Suppose, up to exchanging the two children, that

$$\mathcal{H}_0 = \emptyset \qquad \text{and} \qquad \mathcal{H}_1 = \{H\}.$$

The first child is inconsistent, and we have

$$V_Y(Q) = V_Y(H).$$

There is no need to run the branch solver on H before choosing the next splitting variable.

4.4 Theoretical analysis of variable zeroing

Let Y be the variables in the input system. A path is described by two disjoint subsets of Y: the variables Z set to zero and the variables N declared nonzero. For a system Q on that path, define

$$V_Y(Q; Z, N) = \big\{a \in \mathbb{A}^Y : a_z = 0 \ (z \in Z),\ a_x \neq 0 \ (x \in N), \\ (a, s) \in V(Q) \text{ for some } s\big\}.$$

The quantified coordinate s is omitted when $N = \emptyset$. This notation keeps the zero decisions even though the corresponding variables have already been replaced by zero in the polynomials of Q.

We use the following contract for BranchingWithTimeout. At a path (Z, N), an input Q returns a collection S_Q of solution-set representations and a set T_Q of unresolved branches such that

$$V_Y(Q; Z, N) = \mathrm{Sol}(S_Q) \cup \bigcup_{R \in T_Q} V_Y(R; Z, N). \tag{3}$$

Here $\mathrm{Sol}(S_Q)$ includes the path conditions and is projected onto Y. A timeout places an unprocessed branch in T_Q instead of discarding it. As in Theorem 3, every learned UNSAT core is assumed to be certified.

Lemma 5 (Correctness of a zeroing split) *Let $x \in Y \setminus (Z \cup N)$. Set*

$$Q_0 = Q|_{x=0} \qquad \textit{and} \qquad Q_1 = \textsc{UpdateSaturation}(Q, N \cup \{x\}).$$

Then

$$V_Y(Q; Z, N) = V_Y(Q_0; Z \cup \{x\}, N) \cup V_Y(Q_1; Z, N \cup \{x\}).$$

Proof 9 *Take $a \in V_Y(Q; Z, N)$. If $a_x = 0$, then evaluating a polynomial at a is unchanged when x is first replaced by zero. Hence $a \in V_Y(Q_0; Z \cup \{x\}, N)$. If $a_x \neq 0$, then every variable in $N \cup \{x\}$ is nonzero. Setting*

$$s = \left(a_x \prod_{y \in N} a_y \right)^{-1}$$

satisfies the updated saturation equation, so $a \in V_Y(Q_1; Z, N \cup \{x\})$. The converse follows by the same two observations: substitution is valid under $a_x = 0$, and the saturation equation only records the stated nonzero conditions.

Theorem 5 (Variable zeroing) *Suppose Algorithm 3 returns (S, U, C) on input P. Then*

$$V(P) = \mathrm{Sol}(S) \cup \bigcup_{(R, Z_R, N_R) \in U} V_Y(R; Z_R, N_R).$$

In particular, if $U = \emptyset$, then $\mathrm{Sol}(S) = V(P)$.

Proof 10 *We prove a stronger claim. Suppose the call*

$$\mathrm{SolveByZeroing}(Q, X, Z, N, \tau, C)$$

returns (S', U', C'). Then

$$\begin{aligned} V_Y(Q; Z, N) = \ & \mathrm{Sol}(S') \\ & \cup \bigcup_{(R, Z_R, N_R) \in U'} V_Y(R; Z_R, N_R). \end{aligned} \tag{4}$$

The proof is by induction on $|X|$.

If $X = \emptyset$, the procedure cannot split an unresolved branch. It returns the representations found by BranchingWithTimeout *and places every branch in T_Q into U'. Thus, Equation* (4) *is exactly Equation* (3)*.*

Now suppose $X \neq \emptyset$ and that the claim holds whenever fewer variables are available. Let (S_Q, T_Q) be the result of BranchingWithTimeout *on Q. For each $R \in T_Q$, the procedure chooses some $x \in X$ and recursively processes*

$$R_0 = R|_{x=0} \qquad \text{and} \qquad R_1 = \textsc{UpdateSaturation}(R, N \cup \{x\}).$$

Both calls use the smaller set $X \setminus \{x\}$. Lemma 5 gives

$$V_Y(R; Z, N) = V_Y(R_0; Z \cup \{x\}, N) \cup V_Y(R_1; Z, N \cup \{x\}).$$

Substituting these decompositions into Equation (3) *gives Equation* (4).

Apply the claim to the initial call with $Q = P$ *and* $Z = N = \emptyset$. *Then* $V_Y(P; \emptyset, \emptyset) = V(P)$, *which proves the stated identity. If* U *is empty, the union over* U *is empty and* $\mathrm{Sol}(S) = V(P)$.

Theorem 6 (Termination) *Algorithm 3 terminates in finite time.*

Proof 11 *Each recursive call removes one variable from* X, *so the recursion depth is at most* $|X|$. *Every polynomial system has finitely many factor hitting sets, and the timeout bounds the time spent evaluating each one. The recursion tree is therefore finite.*

The representations returned by the children determine the representation of the hitting set on which the solver timed out. If every child is UNSAT, the timed-out hitting set is also UNSAT. Otherwise, it represents the union of the children's sets.

5 Experiments

5.1 Setup

We implemented Algorithm 2 in Rust 1.97.1 and used Symbolica 1.5 for polynomial arithmetic over the rationals [22]. In all experiments, we solve the branches with a Gröbner-basis computation. The available implementations are M4GB, F5, GVW, F4, and Singular 4.4.1 [16, 10, 11, 9, 8]. A basis containing 1 certifies that the branch is UNSAT. We obtain an UNSAT core by iterative refinement as in Section 5 of [26]. The comparison also includes six direct whole-ideal Gröbner-basis configurations, three decomposition methods from Singular, Macaulay2 1.26.06, and a numerical method from HomotopyContinuation.jl under Julia 1.12.6 [8, 13, 5, 2].

For Algorithm 3, we test M4GB, F5, GVW, Symbolica, and Singular as branch solvers, disable UNSAT-core learning, and impose a three-second limit on each cover computation. We selected this cutoff by trying one, two, three, and five seconds on the hard gated systems. Three seconds is the smallest tested value that lets Singular complete the Katsura-8 residual (about 2.6 seconds); five seconds completes no additional hard system and only delays the subsequent splits.

We ran the benchmark on an Apple M4 Pro with 14 CPU cores and 24 GB of memory. Every table entry is a single run. In the baseline benchmark, every method received 60 seconds per system; the three-second zeroing limit is an inner, per-cover limit within that budget.

For branching and variable zeroing, T is end-to-end wall time and F is the time from the start of the minimal-hitting-set search to its first satisfiable hitting set. Factorization and input processing are therefore included in T but not in F. The complete measurements are stored in the appendix.

5.2 Baseline systems

The baseline has 20 generated systems in six families.

- Signed-distance cycles and affine-factor hypergraphs test many inexpensive hitting sets.
- The overlapping random multiples use seeded dense quadratic factors that recur across equations, with no factor shared by every equation.
- For gated systems, we create a new variable and multiply every equation in the standard Cyclic or Katsura benchmark systems, or in a dense quadratic system, by it [3, 4].
- This produces one easy selector cover and one nonlinear residual cover.

Family	Tested sizes	Variables/equations; hitting sets
Signed-distance cycle	$n = 8, 10, 12$	$8/8, 10/10, 12/12$; $8, 8, 32$
Overlapping random multiples	$(n, p, k) = (4, 6, 2), (6, 8, 3), (8, 10, 3)$	n/n; $4, 16, 28$
Positive affine hypergraph	$(n, m, k) = (8, 6, 4), (10, 8, 5), (12, 10, 6)$	n/m; $63, 239, 833$
Gated Cyclic	orders $5, 6, 7$	$6/5, 7/6, 8/7$; 2 each
Gated Katsura and dense quadratics	Katsura $6, 7, 8$; dense $n = 5, 6, 7$	$8/7, 9/8, 10/9, 6/5, 7/6, 8/7$; 2 each
Square affine hypergraph	$(n, k) = (8, 4), (10, 5)$	n/n; $31, 304$

Table 1: The generated systems used for the baseline benchmark.

The overlapping-random-multiple runs disable interreduction so that the generated factor-incidence pattern reaches the hitting-set enumerator unchanged. All other branching runs use interreduction. Table 2 selects the fastest completed branching backend, the fastest completed zeroing backend observed, and the fastest completed method in each baseline class. A branching or zeroing entry gives T/F in seconds; the other entries give total wall time. "Timeout" means that every tested method in that class timed out. "Partial" means that zeroing returned at least one unresolved branch, and "error" records a failed run rather than a mathematical conclusion.

System	Best branching T/F	Selected zeroing T/F	Best direct GB	Best decomposition	Numerical
Signed distance, $n = 8$	Symb. 0.008/0.000	Symb. 0.008/0.001	Symb. 0.008	facstd 0.190	32.410
Signed distance, $n = 10$	Symb. 0.025/0.002	Symb. 0.025/0.002	Symb. 0.029	facstd 0.430	29.944
Signed distance, $n = 12$	Symb. 0.087/0.002	Symb. 0.091/0.003	Symb. 0.472	primdec 3.056	36.880
Overlapping $(4, 6, 2)$	F5 0.008/0.000	F5 0.009/0.001	Symb. 0.413	facstd 0.265	30.069
Overlapping $(6, 8, 3)$	F5 0.132/0.001	F5 0.132/0.021	timeout	timeout	timeout
Overlapping $(8, 10, 3)$	Singular 2.931/0.010	Singular 2.892/0.152	timeout	timeout	timeout
Positive affine $(8, 6, 4)$	F5 0.008/0.000	Singular 0.181/0.012	timeout	facstd 0.368	50.934
Positive affine $(10, 8, 5)$	F5 0.020/0.000	Symb. 0.060/0.007	timeout	facstd 1.836	timeout
Positive affine $(12, 10, 6)$	F5 0.067/0.000	Singular 9.600/0.078	timeout	facstd 53.318	timeout
Gated Cyclic-5	Symb. 0.009/0.000	Symb. 0.011/0.001	Symb. 0.008	facstd 0.181	35.054
Gated Cyclic-6	Singular 0.045/0.008	Singular 0.043/0.009	Singular 0.309	M2 2.620	37.314
Gated Cyclic-7	timeout	Singular partial 24.118/0.009	timeout	timeout	58.287
Gated Katsura-6	Singular 0.049/0.008	Singular 0.049/0.009	GVW 0.226	facstd 0.233	31.532
Gated Katsura-7	Singular 0.283/0.008	Singular 0.284/0.009	GVW 3.176	facstd 0.539	31.980
Gated Katsura-8	Singular 2.657/0.052	Singular 2.588/0.009	timeout	facstd 3.596	34.264
Gated dense, $n = 5$	Singular 0.029/0.008	Singular 0.029/0.008	M4GB 0.171	facstd 0.198	29.289
Gated dense, $n = 6$	Singular 0.546/0.008	Singular 0.535/0.008	M4GB 4.736	facstd 0.755	30.053
Gated dense, $n = 7$	Singular 42.020/0.017	error	timeout	facstd 48.820	32.261
Square affine $(8, 4)$	Singular 0.470/0.023	Singular 0.445/0.012	timeout	facstd 0.601	timeout
Square affine $(10, 5)$	Symb. 1.174/0.000	Symb. 1.326/0.015	timeout	facstd 2.141	timeout
#systems solved	19	19	10	17	14
#fastest (ties counted)	12	10	2	0	2

Table 2: Best completed baseline results and the fastest completed zeroing backend observed. All times are in seconds.

Branching is a good choice when factorization produces branch ideals that are substantially easier than the expanded input ideal. A large number of hitting sets can still be manageable when branch solving and core checks are cheap; the positive affine family shows that the hitting-set count alone is not decisive. The method is less effective when one cover retains the difficulty of the original nonlinear system, as Cyclic-7 does.

The first-solution measurements support this explanation. On positive affine $(12, 10, 6)$, F rounds to 0.000 seconds despite the 833 hitting sets. On gated Katsura-8, the first satisfiable cover appears in 0.052 seconds, but the full run takes 2.657 seconds. Factorization therefore finds the easy component quickly, while the nonlinear residual controls completion time.

Variable zeroing is intended for a narrower case: a cover has timed out, but branching a variable to zero or nonzero, both simplify it the system. The current benchmark gives limited evidence for that mechanism because none of the completed zeroing runs requires a split. On Cyclic-7, zeroing continues the stalled search and returns a partial result, but it does not complete the residual branch.

5.3 Physical Systems

The system is an order five approximation of the theories that represent a graviton are consistent with experimental results of QFT (quantum field theory) [21, 27].

Solving this system can lead to a better understanding of how quantum gravity may look like [27].

For this larger system, zeroing uses a one-second limit per cover and a 60-second total runtime limit. Ordinary branching uses a 60-second limit per cover and a 65-second external hard stop. Thus, an earlier completed cover can make the reported branching wall time slightly exceed 60 seconds.

Backend	Branching T/F (s)	Branching result	Zeroing T/F (s)	Zeroing result
M4GB	60.793/0.003	partial: 32 SAT, 1 timeout; $d = 14$	*timeout*/1.052	partial: 7 comp., 52 splits, 11 unresolved
F5	61.422/0.003	partial: 32 SAT, 1 timeout; $d = 14$	*timeout*/2.059	partial: 13 comp., 63 splits, 9 unresolved
GVW	61.653/0.003	partial: 32 SAT, 1 timeout; $d = 14$	*timeout*/2.071	partial: 13 comp., 60 splits, 9 unresolved
Symbolica	60.218/0.003	partial: 32 SAT, 1 timeout; $d = 14$	*timeout*/1.051	partial: 7 comp., 56 splits, 12 unresolved
Singular	60.342/0.039	partial: 32 SAT, 1 timeout; $d = 14$	*timeout*/0.566	partial: 1 comp., 44 splits, 12 unresolved
`facstd`	1.067/0.013	complete: 32 SAT, 1 UNSAT; $d = 14$	0.824/0.077	complete: 6 comp., 0 splits, 0 unresolved

Table 3: Physical-system results from the fresh single-run benchmark.

The `facstd` backend is the only tested backend that completes the physical system, and it does so with both ordinary branching and variable zeroing. Every other backend returns only a partial result within the stated limits. Recursive cover refinement produces 33 terminal covers. Each other branching backend finds 32 satisfiable covers, including a dimension-14 component, before timing out on the final nonlinear residual cover.

6 Conclusion

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A Complete experimental results

The tables below give every run summarized in Table 2. Unless stated otherwise, entries are end-to-end wall times in seconds. P marks a partial run and TO an outer timeout. These are single measurements.

A.1 Factor branching

Table 4 reports all five branch-solver configurations. A partial entry found the easy component shown in the main text but left at least one cover unresolved.

System	M4GB	F5	GVW	Symbolica	Singular
Signed distance, $n = 8$	0.651	0.053	0.122	0.008	0.083
Signed distance, $n = 10$	0.046	10.927	45.797	0.025	0.084
Signed distance, $n = 12$	0.182	6.256	1.841	0.087	0.271
Overlapping $(4, 6, 2)$	0.017	0.008	0.011	0.015	0.038
Overlapping $(6, 8, 3)$	1.053	0.132	0.280	0.657	0.201
Overlapping $(8, 10, 3)$	TO	10.937	30.687	TO	2.931
Positive affine $(8, 6, 4)$	0.009	0.008	0.010	0.008	0.497
Positive affine $(10, 8, 5)$	0.021	0.020	0.028	0.020	1.870
Positive affine $(12, 10, 6)$	0.072	0.067	0.096	0.068	6.498
Gated Cyclic-5	0.010	0.010	0.011	0.009	0.022
Gated Cyclic-6	P 60.010	5.960	2.275	P 60.013	0.045
Gated Cyclic-7	P 60.875	P 60.504	P 60.955	P 60.024	P 60.103
Gated Katsura-6	0.366	0.483	1.338	2.540	0.049
Gated Katsura-7	4.487	15.220	42.113	P 60.012	0.283
Gated Katsura-8	54.055	P 60.325	P 60.047	P 60.012	2.657
Gated dense, $n = 5$	0.177	0.071	0.200	0.204	0.029
Gated dense, $n = 6$	4.716	3.194	7.475	21.498	0.546
Gated dense, $n = 7$	P 60.008	P 60.050	P 60.041	P 60.015	42.020
Square affine $(8, 4)$	15.117	TO	TO	3.096	0.470
Square affine $(10, 5)$	14.750	TO	47.561	1.174	2.548

Table 4: End-to-end wall times for every factor-branching configuration.

A.2 Variable zeroing

Table 5 gives the complete output of the fastest completed zeroing configuration used in Table 2. Every backend receives three seconds per cover and a 60-second outer budget, and UNSAT-core learning is disabled. The overlapping-random-multiple

rows disable interreduction, as in the ordinary branching experiment. Here N is the number of recursion nodes, E the number of cover evaluations, O/S the numbers of per-cover timeouts and zeroing splits, and C/U the numbers of returned components and unresolved branches. The time entry is T/F.

System	Backend	T/F	N	E	O/S	C/U	Result
Signed distance, $n = 8$	Symbolica	0.008/0.001	1	8	0/0	8/0	complete
Signed distance, $n = 10$	Symbolica	0.025/0.002	1	8	0/0	8/0	complete
Signed distance, $n = 12$	Symbolica	0.091/0.003	1	32	0/0	32/0	complete
Overlapping $(4, 6, 2)$	F5	0.009/0.001	1	4	0/0	4/0	complete
Overlapping $(6, 8, 3)$	F5	0.132/0.021	1	16	0/0	16/0	complete
Overlapping $(8, 10, 3)$	Singular	2.892/0.152	1	28	0/0	28/0	complete
Positive affine $(8, 6, 4)$	Singular	0.181/0.012	1	4	0/0	4/0	complete
Positive affine $(10, 8, 5)$	Symbolica	0.060/0.007	1	242	0/0	242/0	complete
Positive affine $(12, 10, 6)$	Singular	9.600/0.078	1	511	0/0	511/0	complete
Gated Cyclic-5	Symbolica	0.011/0.001	1	2	0/0	2/0	complete
Gated Cyclic-6	Singular	0.043/0.009	1	2	0/0	2/0	complete
Gated Cyclic-7	Singular	24.118/0.009	15	17	8/7	1/1	partial
Gated Katsura-6	Singular	0.049/0.009	1	2	0/0	2/0	complete
Gated Katsura-7	Singular	0.284/0.009	1	2	0/0	2/0	complete
Gated Katsura-8	Singular	2.588/0.009	1	2	0/0	2/0	complete
Gated dense, $n = 5$	Singular	0.029/0.008	1	2	0/0	2/0	complete
Gated dense, $n = 6$	Singular	0.535/0.008	1	2	0/0	2/0	complete
Gated dense, $n = 7$	Singular	4.797/–	–	–	–	–	error
Square affine $(8, 4)$	Singular	0.445/0.012	1	31	0/0	31/0	complete
Square affine $(10, 5)$	Symbolica	1.326/0.015	1	304	0/0	304/0	complete

Table 5: Selected variable-zeroing result for every system. Times are end-to-end wall time and first-solution latency in seconds.

The dense $n = 7$ run failed while constructing the first recursive child: the expanded Singular program exceeded the operating system's command-line argument limit. This is an implementation error, not an UNSAT result or a timeout. We also tried M4GB, F5, GVW, and Symbolica on this system. These backends execute in-process; after a per-cover timeout their non-cancellable worker threads continued alongside the recursive children until the process was killed under memory pressure. Singular remains preferable for the hard rows because its external worker can be terminated. The Cyclic-7 run exhausted all useful splitting variables on one route and therefore returned one unresolved branch.

A.3 Direct whole-ideal Gröbner bases

System	M4GB	F5	GVW	Symbolica	Singular	Macaulay2
Signed distance, $n = 8$	0.646	0.014	0.010	0.008	0.333	0.570
Signed distance, $n = 10$	0.130	0.221	0.112	0.029	0.194	0.518
Signed distance, $n = 12$	3.266	20.044	8.685	0.472	0.512	1.293
Overlapping $(4, 6, 2)$	1.000	24.865	0.472	0.413	TO	TO
Overlapping $(6, 8, 3)$	TO	TO	TO	TO	TO	TO
Overlapping $(8, 10, 3)$	TO	TO	TO	TO	TO	TO
Positive affine $(8, 6, 4)$	TO	TO	TO	TO	TO	TO
Positive affine $(10, 8, 5)$	TO	TO	TO	TO	TO	TO
Positive affine $(12, 10, 6)$	TO	TO	TO	TO	TO	TO
Gated Cyclic-5	0.012	0.279	0.009	0.008	0.183	0.480
Gated Cyclic-6	TO	TO	1.738	TO	0.309	0.917
Gated Cyclic-7	TO	TO	TO	TO	TO	TO
Gated Katsura-6	0.436	44.961	0.226	5.277	14.430	TO
Gated Katsura-7	5.165	TO	3.176	TO	TO	TO
Gated Katsura-8	TO	TO	TO	TO	TO	TO
Gated dense, $n = 5$	0.171	8.884	0.179	0.200	0.529	TO
Gated dense, $n = 6$	4.736	TO	6.755	17.472	TO	TO
Gated dense, $n = 7$	TO	TO	TO	TO	TO	TO
Square affine $(8, 4)$	TO	TO	TO	TO	TO	TO
Square affine $(10, 5)$	TO	TO	TO	TO	TO	TO

Table 6: End-to-end wall times for every direct whole-ideal Gröbner-basis configuration.

A.4 Decomposition and numerical methods

System	facstd	primdecGTZ	Macaulay2	Homotopy
Signed distance, $n = 8$	0.190	0.320	1.020	32.410
Signed distance, $n = 10$	0.430	0.667	3.224	29.944
Signed distance, $n = 12$	4.800	3.056	25.223	36.880
Overlapping $(4, 6, 2)$	0.265	TO	TO	30.069
Overlapping $(6, 8, 3)$	TO	TO	TO	TO
Overlapping $(8, 10, 3)$	TO	TO	TO	TO
Positive affine $(8, 6, 4)$	0.368	TO	TO	50.934
Positive affine $(10, 8, 5)$	1.836	TO	TO	TO
Positive affine $(12, 10, 6)$	53.318	TO	TO	TO
Gated Cyclic-5	0.181	0.337	0.709	35.054
Gated Cyclic-6	2.729	TO	2.620	37.314
Gated Cyclic-7	TO	TO	TO	58.287
Gated Katsura-6	0.233	TO	TO	31.532
Gated Katsura-7	0.539	TO	TO	31.980
Gated Katsura-8	3.596	TO	TO	34.264
Gated dense, $n = 5$	0.198	TO	TO	29.289
Gated dense, $n = 6$	0.755	TO	TO	30.053
Gated dense, $n = 7$	48.820	TO	TO	32.261
Square affine $(8, 4)$	0.601	TO	TO	TO
Square affine $(10, 5)$	2.141	TO	TO	TO

Table 7: End-to-end wall times for every decomposition and numerical configuration.